DEOVRAT PRASAD

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RESEARCH PROFILE

Theoretical and computational astrophysicist studying how feedback from supermassive black holes shapes the evolution of galaxies, groups, and clusters. Research combines magnetohydrodynamic simulations, high-performance computing, and models of accretion and multiphase gas.

RESEARCH INTERESTS

AGN feedback and accretion; galaxy formation and evolution; feedback processes in galaxies, groups, and clusters; numerical methods for astrophysical fluids.

ACADEMIC APPOINTMENTS

Assistant Professor
Department of Physics, Central University of South Bihar, Gayaji, India | May 2026–present

Postdoctoral Research Associate
Cardiff University, Cardiff, UK | November 2022–June 2025

Postdoctoral Fellow
Michigan State University, East Lansing, Michigan, USA | July 2018–June 2022

Research Associate
Indian Institute of Science, Bengaluru, India | August 2017–June 2018

EDUCATION

PhD in Astronomy and Astrophysics
Indian Institute of Science, Bengaluru, India | June 2018
Thesis: AGN Feedback in Galaxy Clusters: Controlling Cooling Flows in Galaxy Clusters by Momentum-Driven AGN Jets
Advisor: Prateek Sharma | **Funding:** CSIR Junior/Senior Research Fellowship

Integrated MSc in Physics (five-year degree)
UM-DAE Centre for Excellence in Basic Sciences, University of Mumbai, India | 2012
CGPA: 7.9/10 | **Thesis:** Study of Resonance Particles in Quantum Mechanics by Numerical Method
Advisor: Neelmani Mathur, TIFR Mumbai | **Funding:** INSPIRE Fellowship

SELECTED HONORS AND FELLOWSHIPS

- **India Top Cited Author Award**, IOP Publishing and American Astronomical Society (2018; citation period 2015–2017).
- **CSIR NET Fellowship** (2012–2017); 24th All India Rank, National Eligibility Test, June 2012.
- **Certificate of Merit**, National Graduate Physics Examination (2010); national top 1%.
- **INSPIRE Fellowship**, Department of Science and Technology, Government of India (2007–2012).

COMPUTING TIME AWARDS

Study the Evolution of Massive Galaxies with Cold-Mode AGN Feedback at Low Redshift
Supercomputing Wales (Hawk and Sunbird) | Co-PI | 1.0 million CPU hours | November 2022–December 2025

Feedback and Energetics from Magnetized AGN Jets in Galaxy Groups and Clusters
INCITE, Frontier, Oak Ridge Leadership Computing Facility | Co-PI | 630K + 600K GPU node hours | January 2023–December 2024

Probing Galaxy Formation at Low and High Redshifts
NSF XSEDE (Stampede and Expanse) | Co-PI | 1.6 million CPU hours | January–December 2022

Precipitation-Regulated AGN Feedback in Halos from 10^{12} to 10^{15} solar masses
NSF XSEDE (Comet and Expanse) | PI | 2.05 million CPU hours | January 2021–June 2022

Precipitation-Regulated AGN Feedback in Halos from 10^{12} to 10^{15} solar masses
NSF XSEDE (Comet) | PI | 1.23 million CPU hours | July 2019–June 2020

TEACHING

Instructor, Central University of South Bihar | July–November 2026

- Mathematical Physics I, undergraduate course.
- Theory of Relativity, undergraduate course.
- Introduction to Astroparticle Physics, graduate course.

Guest Lecturer, Energy and Gas in Interstellar Space
Graduate course, Cardiff University | Spring 2024 and Spring 2025 | Instructor: Tim Davis

Co-Instructor, Introduction to Astrophysics
Undergraduate course, Cardiff University | Spring 2023 and Spring 2024 | Co-taught with Mathew Smith

Teaching Assistant, Electricity, Magnetism and Optics
Undergraduate course, Indian Institute of Science | January–April 2015 | Instructor: Tarun Deep Saini

Teaching Assistant, Fluids and Plasma
Graduate course, Indian Institute of Science | August–November 2014 | Instructor: Prateek Sharma

STUDENT MENTORING

MSc dissertation mentoring (Ongoing)
Department of Physics, Central University of South Bihar | July 2026–May 2027
Madhu Bala; Kanchan Maurya; Yuvraj Chaitanya; Debjani Das.

Internship mentoring
Department of Physics, Central University of South Bihar | July–November 2026
Shalini Shikha Kumari.

Annual project co-mentor
School of Physics and Astronomy, Cardiff University | September 2024–May 2025
Kobe Richards.

ACRES summer program co-mentor
Department of Computational Mathematics, Science and Engineering, Michigan State University | May–July 2019
Sebastian Lacayo, Florida International University.

ACADEMIC AND ADMINISTRATIVE SERVICE

- **Member**, Research Advisory Committee, Department of Physics, Central University of South Bihar (August 2026–present).
- **Warden**, Boys' Hostel, Central University of South Bihar, Gayaji, India (June 2026–present).
- **Peer review:** Science Advances (2025–2026); The Astrophysical Journal (2025); UKRI-funded computing resources proposal (2024); NASA peer review (2022); Monthly Notices of the Royal Astronomical Society (2018).

COLLABORATIONS

- **XMAGNET / Feedback and Energetics from Magnetized AGN Jets**
  Michigan State University (2023–present)
- **Figuring Out Gas & Galaxies in Enzo (FOGGIE)**
  Space Telescope Science Institute (2019–2022)

COMPUTATIONAL AND TECHNICAL EXPERTISE

Languages: Fortran 77/95, C, C++, Python, MATLAB, MPI.
Simulation codes: ZEUS-MP, PLUTO, Enzo, AthenaPK, AREPO (HD/MHD).
HPC: Experience with XSEDE, NASA, Michigan State University, IISc, and Supercomputing Wales facilities.

- **Enzo:** Co-developer; implemented AGN feedback using the Active Particle formulation.
- **PLUTO:** Co-developer of AGN feedback and Runge–Kutta–Legendre super-time-stepping methods for anisotropic thermal conduction.

PROFESSIONAL DEVELOPMENT

- **Supervised Machine Learning: Regression and Classification**, DeepLearning.AI and Stanford University via Coursera (March 2026).
- **Machine Learning: Advanced Learning Algorithms**, DeepLearning.AI and Stanford University via Coursera (March 2026).

SELECTED PRESENTATIONS

Conference talks

Environmental Effects on Black Hole Feedback Valve in Massive Galaxies
Resolving Galaxy Ecosystems Across All Scales, The Chinese University of Hong Kong, Hong Kong | December 2023

Environmental Effects on Black Hole Feedback Valve in Massive Galaxies
6th ICM Theory and Computation Workshop, Niels Bohr Institute, Copenhagen, Denmark | August 2022

AGN and Stellar Feedback in Galaxy Clusters
ASTRONUM 2019, Paris, France | July 2019

Role of BCG and AGN-Driven Turbulence in Galaxy Cluster Evolution
SnowCluster: The Physics of Galaxy Clusters, Snowbird, Utah, USA | March 2018

AGN Feedback in Galaxy Groups
The Physics of Groups and Galaxy Properties Therein, Institut d'Astrophysique de Paris, France | December 2016

Colloquia and invited presentations

Cool-Core Cycles: From Massive Galaxies to Galaxy Clusters
Raman Research Institute, Bengaluru | January 2026

Environmental Effects on Black Hole Feedback Valve in Massive Galaxies
Department of Physics, Indian Institute of Science, Bengaluru | October 2022

Cool-Core Cycles: AGN Feedback in Galaxy Groups and Clusters
Inter-University Centre for Astronomy and Astrophysics, Pune | September 2022

Cool-Core Cycles: AGN Feedback in Galaxy Groups and Clusters
Department of Astronomy and Astrophysics, Tata Institute of Fundamental Research, Mumbai | September 2022

Environmental Dependence of Self-Regulating Black Hole Feedback in Massive Galaxies
HALO 2021, KITP | Short video presentation | January–March 2021

Environmental Effects on Black Hole Feedback Valve in Massive Galaxies
Astro Group Meeting, Center for Computational Astrophysics, Flatiron Institute | October 2020

The Feedback Loop: Controlling Cooling Flows in Galaxy Clusters
Astronomy Colloquium, University of Notre Dame, USA | December 2018

REFERENCES

Mark Voit
Professor, Department of Physics and Astronomy, Michigan State University
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Brian O'Shea
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Associate Professor, Department of Physics and Astronomy, Indian Institute of Science
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Distinguished Professor, Department of Physics and Astronomy, University of Victoria
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Freeke van de Voort
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PUBLICATIONS

Refereed articles — lead author

XMAGNET: Kinetic, Thermal and Magnetic AGN Feedback in Massive Galaxies at Halo Masses $\approx 10^{13.5}$ solar masses
Deovrat Prasad, Philipp Grete, Brian O'Shea, Forrest Glines, Mark Voit, Freeke van de Voort, Martin Fournier, and Ben Wibking. 2026, MNRAS, 545, 1–18.

The Case for Hot-Mode Accretion in Abell 2029
Deovrat Prasad, Mark Voit, and Brian O'Shea. 2024, MNRAS, 531, 259.

Atmospheric Circulation in Simulations of the AGN–CGM Connection at Halo Masses $\approx 10^{13.5}$ solar masses
Deovrat Prasad, Mark Voit, and Brian O'Shea. 2022, ApJ, 932, 18.

Environmental Dependence of Self-Regulating Black-Hole Feedback in Massive Galaxies
Deovrat Prasad, Mark Voit, Brian O'Shea, and Forrest Glines. 2020, ApJ, 905, 50.

Cool-Core Cycles and Phoenix
Deovrat Prasad, Prateek Sharma, Arif Babul, Mark Voit, and Brian O'Shea. 2020, MNRAS, 495, 594.

Cool-Core Clusters: Role of BCG, Star Formation, and AGN-Driven Turbulence
Deovrat Prasad, Prateek Sharma, and Arif Babul. 2018, ApJ, 863, 62.

AGN Jets Drive Stochastic Cold Accretion in Cluster Cores
Deovrat Prasad, Prateek Sharma, and Arif Babul. 2017, MNRAS, 471, 1531.

Cool-Core Cycles: Cold Gas and AGN Jet Feedback in Cluster Cores
Deovrat Prasad, Prateek Sharma, and Arif Babul. 2015, ApJ, 811, 108.

Refereed articles — contributing author

XMAGNET – Stir before Serving: A Lagrangian Perspective on Mixing-Driven Condensation in the Intracluster Medium
M. Fournier, P. Grete, M. Brüggen, B. W. O'Shea, G. M. Voit, B. D. Wibking, and D. Prasad. 2026. Accepted for publication in Astronomy & Astrophysics. arXiv:2605.00563.

XMAGNET: Velocity Structure Functions of AGN-Driven Turbulence in the Multiphase Intracluster Medium
Martin Fournier, Philipp Grete, Marcus Brüggen, Brian W. O'Shea, Deovrat Prasad, Benjamin D. Wibking, Forrest W. Glines, and Rajshekhar Mohapatra. 2025, A&A;, 698, A121.

The XMAGNET Exascale MHD Simulations of SMBH Feedback in Galaxy Groups and Clusters: Overview and Preliminary Cluster Results
Philipp Grete, Brian W. O'Shea, Forrest W. Glines, Deovrat Prasad, Benjamin D. Wibking, Martin Fournier, Marcus Brüggen, and G. Mark Voit. 2025, ApJ, 988, 155.

A Black-Hole Feedback Valve in Massive Galaxies
G. Mark Voit, Greg L. Bryan, Deovrat Prasad, Rachel Frisbie, Yuan Li, Megan Donahue, Brian W. O'Shea, Ming Sun, and Norbert Werner. 2020, ApJ, 899, 70.

Figuring Out Gas & Galaxies in Enzo (FOGGIE). IV. The Stochasticity of Ram Pressure Stripping in Galactic Halos
Raymond C. Simons, Molly S. Peeples, Jason Tumlinson, Brian W. O'Shea, Britton D. Smith, Lauren Corlies, Cassandra Lochhaas, Yong Zheng, Ramona Augustin, Deovrat Prasad, Gregory F. Snyder, and Erik Tollerud. 2020, ApJ, 905, 167.

Scalable Explicit Implementation of Anisotropic Diffusion with Runge–Kutta–Legendre Super-Time-Stepping
Bhargava Vaidya, Deovrat Prasad, Andrea Mignone, Prateek Sharma, and Luca Rickler. 2017, MNRAS, 472, 3147.

Non-refereed publication

Circumgalactic Gas and the Precipitation Limit (Astro2020 Science White Paper)
G. M. Voit, A. Babul, Iu. Babyk, G. L. Bryan, H.-W. Chen, M. Donahue, D. Fielding, M. Gaspari, Y. Li, M. McDonald, B. W. O'Shea, D. Prasad, P. Sharma, M. Sun, G. Tremblay, J. Werk, N. Werner, and F. Zahedy. 2019, arXiv:1903.11212.

Complete publication list: NASA ADS public library